

PrimeCore Minerals

Site-to-Site VPN

Implementation and

Validation

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Date: October 4th, 2026

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1. Introduction

This technical artefact documents the implementation and testing of the site-to-site VPN used to connect the PrimeCore Minerals Brisbane HQ and Mine Site A networks. The VPN was configured using pfSense at both Brisbane HQ and Mine Site A to provide a secure connection between the two locations. Only the internal networks that require communication between the sites are included in the VPN, while the pfSense firewall rules are used to control what traffic is allowed to pass through.

The VPN was tested to confirm that communication between the two locations was working as intended. This included testing traffic between the Employee and Server networks, confirming that Mine Site A firewall logs could be sent through the VPN to HQ-MONITOR, and using a WAN packet capture to verify that IKE and ESP traffic was travelling between both pfSense firewalls.

This artefact explains the VPN design, the Phase 1 and Phase 2 configuration, the firewall rules used to control traffic between the two sites, and the testing that was completed. It also covers the problems found during testing, the changes that were made to fix them, and the current limitations of the VPN setup.

2. Purpose and Scope

The purpose of the site-to-site VPN is to securely connect Brisbane HQ and Mine Site A while still keeping the existing VLANs and firewall controls separated at each location.

The VPN was implemented to:

- Provide a secure connection between Brisbane HQ and Mine Site A
- Allow approved Employee and Server network traffic to move between both locations
- Support central services hosted at Brisbane HQ
- Allow Mine Site A firewall logs to be sent back to HQ-MONITOR
- Keep inter-site traffic controlled by pfSense firewall rules
- Provide a network design that could be adapted for additional mine sites in the future
- Provide evidence that inter-site traffic is travelling through the IPsec tunnel

The current VPN setup connects Brisbane HQ and Mine Site A. Automatic VPN failover and a second redundant VPN connection were not implemented in the final prototype, so these would be considered as future improvements.

3. VPN Network Design

The pfSense firewalls at Brisbane HQ and Mine Site A are connected through the VPN. HQ-FW uses the WAN address 172.16.100.2/30, while SITE-A-FW uses 172.16.101.2/30. These two firewall addresses are used as the main VPN endpoints for the connection between both locations.

The traffic between the two firewalls passes through the ISP networks shown below:

Device	Interface Address	Purpose
HQ-FW	172.16.100.2/30	Brisbane HQ VPN endpoint
HQ-ISP	172.16.100.1/30	HQ ISP connection
HQ-ISP transit	172.16.200.1/30	Link between both ISP networks
SITE-A-ISP transit	172.16.200.2/30	Link between both ISP networks
SITE-A-ISP	172.16.101.1/30	Mine Site A ISP connection
SITE-A-FW	172.16.101.2/30	Mine Site A VPN endpoint

This setup provides the path used by the site-to-site VPN between Brisbane HQ and Mine Site A. The traffic travelling between 172.16.100.2 and 172.16.101.2 is encrypted by the VPN, while the internal networks at each location continue to be controlled separately through their own pfSense firewall.

The VPN is set up to connect the Employee and Server networks at both locations. Brisbane HQ uses 192.168.10.0/24 for Employees and 192.168.90.0/24 for Servers, while Mine Site A uses 192.168.110.0/24 and 192.168.190.0/24. These networks are added to the VPN through the Phase 2 configuration, while the other VLANs are separated from the tunnel unless they need to be added later.

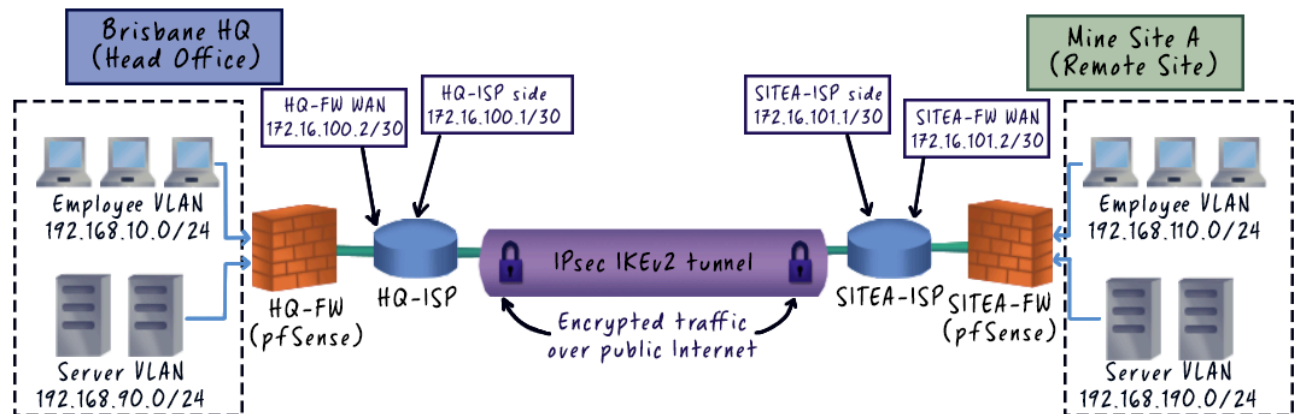


Figure 1: Site-to-site VPN diagram of secure connection between Brisbane HQ and Mine Site A.

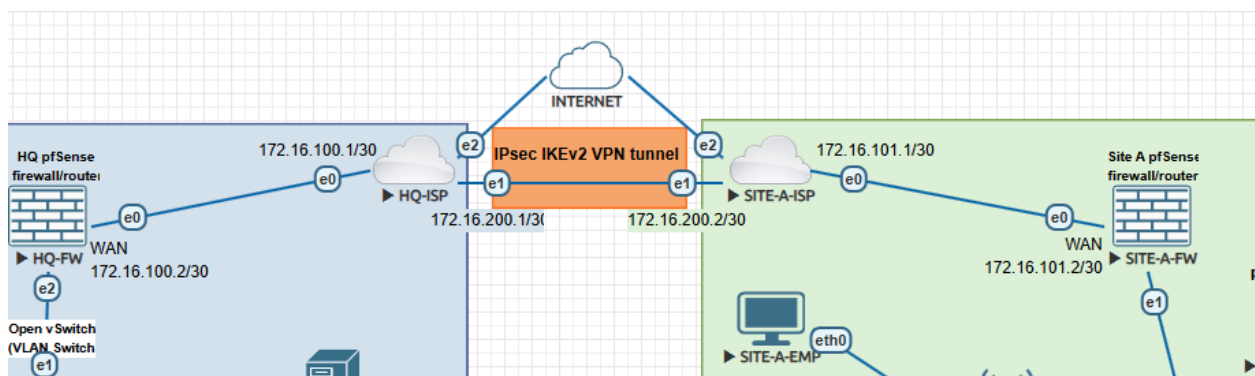


Figure 2: EVE-NG network topology showing the routed ISP path between HQ-FW and SITE-A-FW used by the site-to-site VPN.

4. VPN Configuration

pfSense is used at both Brisbane HQ and Mine Site A to create the VPN connection between the two locations. HQ-FW connects to SITE-A-FW using the address 172.16.101.2 and connects back to HQ-FW using the address 172.16.100.2. The connection uses IKEv2/IPsec to create the secure tunnel between both firewalls.

Location	Firewall	Local WAN	Remote Peer
Brisbane HQ	HQ-FW	172.16.100.2	172.16.101.2
Mine Site A	SITE-A-FW	172.16.101.2	172.16.100.2

The VPN uses a pre-shared key to authenticate the connection between both firewalls. For security I have refrained from including the key value from the artefact and it has been redacted from screenshots.

IPsec Tunnels										
	ID	IKE	Remote Gateway		Auth/Mode	P1 Protocol	P1 Transforms	P1 DH-Group	P1 Description	Actions
☐	Disable	1	V2	WAN	Mutual PSK	AES (256 bits)	SHA256	14 (2048 bit)	HQ to SiteA IKEv2	
				172.16.101.2	-					
	ID	Mode	Local Subnet	Remote Subnet	P2 Protocol	P2 Transforms	P2 Auth Methods	Description	P2 actions	
☐	Disable	1	tunnel	192.168.10.0/24	192.168.110.0/24	ESP	AES (256 bits)	SHA256	HQ employee to siteA employee	
☐	Disable	2	tunnel	192.168.90.0/24	192.168.190.0/24	ESP	AES (256 bits)	SHA256	HQ server to Site A Server	
	+ Add P2									

Figure 3: HQ-FW IPsec tunnel Phase 1 connection and Employee and Server Phase 2 connections to Mine Site A.

IPsec Tunnels										
	ID	IKE	Remote Gateway	Auth/Mode	P1 Protocol	P1 Transforms	P1 DH-Group	P1 Description	Actions	
☐	Disable	1	V2	WAN 172.16.100.2	Mutual PSK -	AES (256 bits)	SHA256	14 (2048 bit)	SiteA to HQ IKEV2	
	ID	Mode	Local Subnet	Remote Subnet	P2 Protocol	P2 Transforms	P2 Auth Methods	Description	P2 actions	
☐	Disable	1	tunnel	192.168.110.0/24	192.168.10.0/24	ESP	AES (256 bits)	SHA256	Site A Employee to HQ employee	
☐	Disable	2	tunnel	192.168.190.0/24	192.168.90.0/24	ESP	AES (256 bits)	SHA256	Site A server to HQ server	
Add P2										

Figure 4: SITE-A-FW IPsec tunnel showing the Phase 1 and Phase 2 connections to Brisbane HQ.

4.1 Phase 1 Configuration

Phase 1 is used to create the connection between the two pfSense firewalls. Brisbane HQ and Mine Site A use IKEv2 over IPv4, with the connection running through the WAN interface. HQ-FW uses 172.16.101.2 as the remote gateway, with SITE-A-FW using 172.16.100.2.

The connection is authenticated using a shared pre-shared key on both firewalls. Each side is identified using its WAN address, with the opposite firewall set as the peer. For Phase 1, the VPN uses AES-256 for encryption, SHA256 for hashing and Diffie-Hellman Group 14.

General Information			
Description	HQ to SiteA IKEv2 A description may be entered here for administrative reference (not parsed).		
Disabled	☐ Set this option to disable this phase1 without removing it from the list.		
IKE ID	1		

IKE Endpoint Configuration			
Key Exchange version	IKEv2 Select the Internet Key Exchange protocol version to be used. Auto uses IKEv2 when initiator, and accepts either IKEv1 or IKEv2 as responder.		
Internet Protocol	IPv4 Select the Internet Protocol family.		
Interface	WAN Select the interface for the local endpoint of this phase1 entry.		
Remote Gateway	172.16.101.2 Enter the public IP address or host name of the remote gateway. ⓘ		

Phase 1 Proposal (Authentication)			
Authentication Method	Mutual PSK Must match the setting chosen on the remote side.		
My identifier	IP address	172.16.100.2	
Peer identifier	IP address	172.16.101.2	
Pre-Shared Key	 Enter the Pre-Shared Key string. This key must match on both peers. This key should be long and random to protect the tunnel and its contents. A weak Pre-Shared Key can lead to a tunnel compromise. Generate new Pre-Shared Key		

Phase 1 Proposal (Encryption Algorithm)					
Encryption Algorithm	AES	256 bits	SHA256	14 (2048 bit)	Delete
	Algorithm	Key length	Hash	DH Group	
Note: SHA1 and DH groups 1, 2, 5, 22, 23, and 24 provide weak security and should be avoided.					
Add Algorithm	+ Add Algorithm				

Figure 5: HQ-FW IPsec Phase 1 configuration showing the IKEv2 settings, Mine Site A remote peer and authentication configuration.

General Information	
Description	SiteA to HQ IKEV2 A description may be entered here for administrative reference (not parsed).
Disabled	☐ Set this option to disable this phase1 without removing it from the list.
IKE ID	1
IKE Endpoint Configuration	
Key Exchange version	IKEv2 Select the Internet Key Exchange protocol version to be used. Auto uses IKEv2 when initiator, and accepts either IKEv1 or IKEv2 as responder.
Internet Protocol	IPv4 Select the Internet Protocol family.
Interface	WAN Select the interface for the local endpoint of this phase1 entry.
Remote Gateway	172.16.100.2 Enter the public IP address or host name of the remote gateway. 
Phase 1 Proposal (Authentication)	
Authentication Method	Mutual PSK Must match the setting chosen on the remote side.
My identifier	IP address 172.16.101.2
Peer identifier	IP address 172.16.100.2
Pre-Shared Key	Enter the Pre-Shared Key string. This key must match on both peers. This key should be long and random to protect the tunnel and its contents. A weak Pre-Shared Key can lead to a tunnel compromise.  Generate new Pre-Shared Key
Phase 1 Proposal (Encryption Algorithm)	
Encryption Algorithm	AES 256 bits SHA256 14 (2048 bit)  Delete Algorithm Key length Hash DH Group
Note: SHA1 and DH groups 1, 2, 5, 22, 23, and 24 provide weak security and should be avoided.	
Add Algorithm	 Add Algorithm

Figure 6: SITE-A-FW Phase 1 configuration showing the IKEv2, authentication and encryption settings used for the connection to Brisbane HQ.

4.2 Phase 2 Configuration

Phase 2 is used to choose which internal networks at Brisbane HQ and Mine Site A are able to use the VPN. Separate connections were set up for the Employee and Server networks so that only the networks that need to communicate between both locations are included in the tunnel.

For the Employee networks, Brisbane HQ uses 192.168.10.0/24, while Mine Site A uses 192.168.110.0/24. The Server networks are set up in a similar way, with Brisbane HQ using 192.168.90.0/24 and Mine Site A using 192.168.190.0/24. These Phase 2 settings are basically used to decide which internal networks are included in the VPN. The pfSense firewall rules then provide another level of control by deciding what traffic is actually allowed to pass between the two locations.

Phase 2 Connections	Brisbane HQ Network	Mine Site A Network
Employee	192.168.10.0/24	192.168.110.0/24
Server	192.168.90.0/24	192.168.190.0/24

The Phase 2 connections use ESP with AES-256 encryption and SHA256 for hashing. PFS Group 14 is also used as part of the Phase 2 configuration. The same settings are configured on both sides of the VPN so that the networks can communicate through the tunnel.

General Information	
Description	HQ employee to siteA employee A description may be entered here for administrative reference (not parsed).
Disabled	☐ Disable this phase 2 entry without removing it from the list.
Mode	Tunnel IPv4
Phase 1	HQ to SiteA IKEv2 (IKE ID 1)
P2 reqid	1
Networks	
Local Network	Network: 192.168.10.0 / 24 Type: Address Local network component of this IPsec security association.
NAT/BINAT translation	None / 0 Type: Address If NAT/BINAT is required on this network specify the address to be translated
Remote Network	Network: 192.168.110.0 / 24 Type: Address Remote network component of this IPsec security association.
Phase 2 Proposal (SA/Key Exchange)	
Protocol	ESP Encapsulating Security Payload (ESP) performs encryption and authentication, Authentication Header (AH) is authentication only.
Encryption Algorithms	☒ AES 256 bits ☐ AES128-GCM Auto ☐ AES192-GCM Auto ☐ AES256-GCM Auto ☐ CHACHA20-POLY1305
Hash Algorithms	☐ SHA1 ☒ SHA256 ☐ SHA384 ☐ SHA512 ☐ AES-XCBC Note: Hash is ignored with GCM algorithms. SHA1 provides weak security and should be avoided.
PFS key group	14 (2048 bit) Note: Groups 1, 2, 5, 22, 23, and 24 provide weak security and should be avoided.

Figure 7: HQ-FW Phase 2 Employee network connection between Brisbane HQ and Mine Site A.

General Information	
Description	HQ server to Site A Server A description may be entered here for administrative reference (not parsed).
Disabled	☐ Disable this phase 2 entry without removing it from the list.
Mode	Tunnel IPv4
Phase 1	HQ to SiteA IKEv2 (IKE ID 1)
P2 reqid	2
Networks	
Local Network	Network: 192.168.90.0 / 24 Type: Address Local network component of this IPsec security association.
NAT/BINAT translation	None / 0 Type: Address If NAT/BINAT is required on this network specify the address to be translated
Remote Network	Network: 192.168.190.0 / 24 Type: Address Remote network component of this IPsec security association.
Phase 2 Proposal (SA/Key Exchange)	
Protocol	ESP Encapsulating Security Payload (ESP) performs encryption and authentication, Authentication Header (AH) is authentication only.
Encryption Algorithms	☒ AES 256 bits ☐ AES128-GCM Auto ☐ AES192-GCM Auto ☐ AES256-GCM Auto ☐ CHACHA20-POLY1305
Hash Algorithms	☐ SHA1 ☒ SHA256 ☐ SHA384 ☐ SHA512 ☐ AES-XCBC Note: Hash is ignored with GCM algorithms. SHA1 provides weak security and should be avoided.
PFS key group	14 (2048 bit) Note: Groups 1, 2, 5, 22, 23, and 24 provide weak security and should be avoided.

Figure 8: HQ-FW Phase 2 configuration showing the Server network connection between Brisbane HQ and Mine Site A.

5. VPN Firewall Rules

Firewall rules were required for the VPN on both pfSense firewalls and were used to control traffic passing between the Brisbane HQ and Mine Site A and how the connection is established. The WAN interfaces are used for the traffic needed to establish the VPN, while the IPsec rules are used to control traffic after it reaches the other firewall.

5.1 WAN VPN Rules

The WAN rules were set up on both pfSense firewalls so the VPN could connect between Brisbane HQ and Mine Site A. On HQ-FW, the rules allow the required VPN traffic from 172.16.101.2, while SITE-A-FW allows the matching traffic from 172.16.100.2. The connection uses UDP port 500 for IKE, UDP port 4500 for NAT-T and ESP for the IPsec traffic. These rules are kept specific to the VPN so the WAN interfaces are not opened up more than they need to be.

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	0/10 KiB	*	Reserved Not assigned by IANA	*	*	*	*	*		Block bogon networks	
☐	0/1 KiB	IPv4 ICMP	172.16.101.2	*	This Firewall (self)	*	*	none			
☐	0/3.68 MiB	IPv4 UDP	172.16.101.2	*	WAN address	IPSEC_PORTS	*	none			
☐	1/255.32 MiB	IPv4 ESP	172.16.101.2	*	WAN address	*	*	none		Allow site a IPsec ESP	
☐	0/0 B	IPv4 ICMP	EMPLOYEE_VLAN subnets	*	192.168.110.0/24	*	*	none		allow hq employee icmp to site a	

Figure 9: HQ-FW WAN firewall rules allowing the traffic required for the site-to-site VPN.

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	0/0 B	*	Reserved Not assigned by IANA	*	*	*	*	*		Block bogon networks	
☐	0/0 B	IPv4 ICMP	172.16.100.2	*	This Firewall (self)	*	*	none		Allow HQ_FW WAN ping	
☐	0/108 B	IPv4 UDP	172.16.100.2	*	WAN address	IPSEC_PORTS	*	none		Allow HQ IKE and NAT-T	
☐	0/0 B	IPv4 ESP	172.16.100.2	*	WAN address	*	*	none		Allow HQ IPsec ESP	

Figure 10: SITE-A-FW WAN firewall rules allowing the traffic required for the site-to-site VPN.

5.2 IPsec Firewall Rules

After the traffic comes through the VPN, the IPsec firewall rules are used to control where that traffic is allowed to go. Just because a network is included in the Phase 2 setup does not mean it can freely access everything at the other location. The rules allow the Employee networks to communicate for VPN testing and also allow the required traffic between the Server networks. HQ-FW also has a separate rule that allows Mine Site A firewall logs to be sent to HQ-MONITOR at 192.168.90.60 using UDP port 514.

Firewall	Source	Destination	Service	Purpose
HQ-FW	192.168.110.0/24	192.168.10.0/24	ICMP	Site A Employee to HQ Employee testing
HQ-FW	192.168.190.0/24	192.168.90.60	UDP 514	Site A firewall logs to HQ-MONITOR
HQ-FW	192.168.190.0/24	192.168.90.60	ICMP	Site A Server to HQ Server testing
SITE-A-FW	192.168.10.0/24	192.168.110.0/24	ICMP	HQ Employee to Site A Employee testing
SITE-A-FW	192.168.90.0/24	192.168.190.0/24	ICMP	HQ Server to Site A Server testing

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 0/1008 B	IPv4 ICMP any	192.168.110.0/24	*	192.168.10.0/24	*	*	none		Allow site a employee ICMP to HQ Employee	
☐	✓ 1/189.09 MiB	IPv4 UDP	192.168.190.0/24	*	192.168.90.60	514 (Syslog)	*	none		Allow site a logging to HQ monitor	
☐	✓ 0/672 B	IPv4 ICMP any	192.168.190.0/24	*	192.168.90.0/24	*	*	none		Allow Site A Server to HQ Server ICMP	

Figure 11: HQ-FW IPsec firewall rules controlling traffic received from Mine Site A.

Rules (Drag to Change Order)											
☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 0/672 B	IPv4 ICMP any	192.168.10.0/24	*	192.168.110.0/24	*	*	none		Allow HQ employee icmp to site A employee	
☐	✓ 0/0 B	IPv4 ICMP any	192.168.90.0/24	*	192.168.190.0/24	*	*	none		Allow HQ server icmp to site a server	

Figure 12: SITE-A-FW IPsec firewall rules controlling traffic received from Brisbane HQ.

6. Testing and Validation

A few different strategies were used to test the VPN and to make sure the connection between Brisbane HQ and Mine Site A was working properly. This included checking the traffic travelling between both WAN addresses, testing communication between the internal networks and confirming that Mine Site A firewall logs could be sent through the VPN to HQ-MONITOR.

6.1 WAN Packet Capture

A packet capture was taken on the HQ-ISP connection to check the traffic moving between the two VPN endpoints. Wireshark was used to view the traffic between HQ-FW at 172.16.100.2 and SITE-A-FW at 172.16.101.2. The capture showed IKE and ESP traffic travelling between both firewall addresses, which helped confirm that the IPsec tunnel was active and the VPN traffic was passing between Brisbane HQ and Mine Site A.

IKE is used to set up and manage the secure VPN connection, while ESP is used to carry the encrypted traffic through the tunnel.

No.	Time	Source	Destination	Protocol	Length	Info
320	51.553973	172.16.101.2	172.16.100.2	ESP	330	ESP (SPI=0xcaef27b0)
327	52.559626	172.16.101.2	172.16.100.2	ESP	298	ESP (SPI=0xcaef27b0)
334	53.559519	172.16.101.2	172.16.100.2	ESP	298	ESP (SPI=0xcaef27b0)
339	54.559793	172.16.101.2	172.16.100.2	ESP	298	ESP (SPI=0xcaef27b0)
344	55.559523	172.16.101.2	172.16.100.2	ESP	298	ESP (SPI=0xcaef27b0)
349	56.559529	172.16.101.2	172.16.100.2	ESP	298	ESP (SPI=0xcaef27b0)
360	58.559625	172.16.101.2	172.16.100.2	ESP	298	ESP (SPI=0xcaef27b0)
369	60.467003	172.16.100.2	172.16.101.2	ISAKMP	122	INFORMATIONAL MID=1955 Initiator Request
370	60.469021	172.16.101.2	172.16.100.2	ISAKMP	122	INFORMATIONAL MID=1955 Responder Response
377	61.559648	172.16.101.2	172.16.100.2	ESP	330	ESP (SPI=0xcaef27b0)
382	62.559536	172.16.101.2	172.16.100.2	ESP	298	ESP (SPI=0xcaef27b0)
389	63.559629	172.16.101.2	172.16.100.2	ESP	298	ESP (SPI=0xcaef27b0)
396	64.560251	172.16.101.2	172.16.100.2	ESP	298	ESP (SPI=0xcaef27b0)
401	65.558963	172.16.101.2	172.16.100.2	ESP	298	ESP (SPI=0xcaef27b0)

Figure 13: Wireshark packet capture showing IKE and ESP traffic between HQ-FW and SITE-A-FW.

6.2 Mine Site A Logging Test

The VPN was tested to verify Site A firewall logs can consistently pass through the VPN tunnel. SITE-A-FW sends its firewall logs from the Site A Server network to HQ-MONITOR at 192.168.90.60 using UDP port 514.

To verify the VPN was handling them correctly, HQ-MONITOR was checked to see if firewall logs from SITE-A-FW were being received correctly. The `/var/log/sitea-pfsense.log` file showed new events coming from the Site A firewall, which proved the logs had successfully travelled through the VPN and reached the HQ monitoring server.

```
,59,192.168.199.40,8.8.8.8,36110,53,39
2026-10-05T06:55:42+00:00 192.168.190.1 filterlog[47293]: 1,6,,1000000103,vtnet1.99,match,block,in,4,0x0,,64,35941,0,DF,6,tcp,60
,192.168.199.111,8.8.8.8,36112,53,0,S,992703702,,64240,,mss;sackOK;TS;nop:wscale
2026-10-05T06:55:43+00:00 192.168.190.1 filterlog[47293]: 1,6,,1000000103,vtnet1.99,match,block,in,4,0x0,,64,35942,0,DF,6,tcp,60
,192.168.199.111,8.8.8.8,36112,53,0,S,992703702,,64240,,mss;sackOK;TS;nop:wscale
2026-10-05T06:55:44+00:00 192.168.190.1 filterlog[47293]: 1,6,,1000000103,vtnet1.99,match,block,in,4,0x0,,64,35943,0,DF,6,tcp,60
,192.168.199.111,8.8.8.8,36112,53,0,S,992703702,,64240,,mss;sackOK;TS;nop:wscale
2026-10-05T06:55:45+00:00 192.168.190.1 filterlog[47293]: 1,6,,1000000103,vtnet1.99,match,block,in,4,0x0,,64,35944,0,DF,6,tcp,60
,192.168.199.111,8.8.8.8,36112,53,0,S,992703702,,64240,,mss;sackOK;TS;nop:wscale
2026-10-05T06:55:46+00:00 192.168.190.1 filterlog[47293]: 1,6,,1000000103,vtnet1.99,match,block,in,4,0x0,,64,35945,0,DF,6,tcp,60
,192.168.199.111,8.8.8.8,36112,53,0,S,992703702,,64240,,mss;sackOK;TS;nop:wscale
2026-10-05T06:55:46+00:00 192.168.190.1 filterlog[47293]: 1,6,,1000000103,vtnet1.99,match,block,in,4,0x0,,64,58260,0,DF,6,tcp,64
,192.168.199.40,192.168.199.1,40548,53,0,S,3828597401,,64240,,mss;sackOK;TS;nop:wscale:Unknown Option %u:nop:nop
```

Figure 14: HQ-MONITOR showing firewall events received from SITE-A-FW in `/var/log/sitea-pfsense.log`.

6.3 Internal Network Connectivity Test

Connectivity between Brisbane HQ and Mine Site A was tested to make sure the internal networks could communicate through the VPN. A ping was sent from the HQ employee network to the Mine Site A employee host at 192.168.110.200.

The ping returned replies from the Site A host, confirming that the HQ employee network was able to communicate with Mine Site A through the VPN. This showed that the

connection between both sites was working correctly and traffic was being routed through the tunnel.

```
labadmin@HQ-EMP-LAN:~$ ip -br addr
lo                UNKNOWN          127.0.0.1/8 ::1/128
ens3              UP                192.168.10.72/24 fe80::250:ff:fe00:700/64
labadmin@HQ-EMP-LAN:~$ ping -c 4 192.168.110.200
PING 192.168.110.200 (192.168.110.200) 56(84) bytes of data.
64 bytes from 192.168.110.200: icmp_seq=1 ttl=62 time=13.5 ms
64 bytes from 192.168.110.200: icmp_seq=2 ttl=62 time=5.78 ms
64 bytes from 192.168.110.200: icmp_seq=3 ttl=62 time=5.37 ms
64 bytes from 192.168.110.200: icmp_seq=4 ttl=62 time=6.06 ms

--- 192.168.110.200 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 5.374/7.671/13.468/3.355 ms
labadmin@HQ-EMP-LAN:~$
```

Figure 15: Successful ping from the Brisbane HQ employee network to the Mine Site A employee.

7. Security Considerations

The VPN was configured so that connecting Brisbane HQ and Mine Site A did not automatically give both sites complete access to each other. Only the Employee and Server networks that needed to communicate were included in the Phase 2 configuration.

Firewall rules were then used to control what traffic was allowed after it reached each pfSense firewall. This meant that even if a network was part of the VPN, it still could not access every device or service at the other site.

The WAN rules were also kept limited to the traffic required for the VPN, including IKE and IPsec traffic. The pre-shared key used to authenticate the VPN was not included in the artefact or screenshots to avoid exposing sensitive information.

The Site A logging rule was also restricted to the required service. UDP port 514 was allowed so SITE-A-FW could send logs to HQ-MONITOR without giving the Site A Server network broader access to the Brisbane HQ Server network.

8. Troubleshooting and Changes

8.1 Employee VPN Rule

One Employee-to-Site-A firewall rule was initially placed on the WAN interface, which stopped the traffic from passing through pfSense correctly. It was moved to the Employee VLAN interface, which allowed the traffic to pass through the VPN correctly.

8.2 Missing HQ IPsec Rule

While testing, traffic from the Mine Site A Server network to HQ-MONITOR was being blocked. An extra IPsec rule was added on HQ-FW to allow the required traffic. After this addition, the connection worked, and Site A firewall logs could reach HQ-MONITOR.

9. Limitations and Future Improvements

The current VPN setup only connects Brisbane HQ and Mine Site A, so the remaining mine sites would still need their own VPN configuration and testing before they could be added to the network.

Automatic VPN failover and a second backup tunnel were not implemented in the final prototype. These would be useful future improvements to help keep communication available if one connection failed.

The VPN was also tested within the EVE-NG environment rather than across real production internet connections. Further testing could include checking performance, latency and how quickly the tunnel recovers after a connection is interrupted.



10. Conclusion

The site-to-site VPN was set up between Brisbane HQ and Mine Site A using pfSense and IKEv2/IPsec. Testing verified that traffic could move between the internal networks, Site A firewall logs were reaching HQ-MONITOR, and IKE and ESP traffic was passing between both VPN endpoints.

Firewall rules were also used to control what traffic could pass between the two locations. The final setup worked as intended and provides a base that could later be extended to the remaining mine sites.